**Imperial College London**

**Department of Electrical & Electronic Engineering**

**ELEC50001 Circuits & Systems – Mid-term Lab Oral 2026**

The mid-term lab oral in the form of a 15-minutes oral examination conducted in person in Level 1 Teaching Lab. This will take place on Monday 2nd or Tuesday 3rd of November 2026. The purpose of this mid-term lab oral is to assess your level of attainment in the intended learning outcomes of **Lab 1 and Lab 2 only**. A schedule for the oral examination is attached. **This is a formal assessment.** If you are unable to attend and wish to swap with someone else, you MUST find a willing person to swap with, and seek my agreement ahead of time via email. The mid-term Lab Oral is worth 15% of the module's marks.

**Logbook**

You must **bring your logbook** (electronics or paper) with you to the Oral Examination. Although your logbook will NOT be formally "marked", your examiner will nevertheless assess how effective you have kept a log of the experiment and whether you have used your logbook to help you learn through planning and reflection. Your examiner will also be asking questions that can **only** be answered by referring to the logbook.

**Format of the Oral** – You must turn up to the Level 1 Lab at least 5 minutes before your allocated time. Your Examiner will call you over to their station when it is your turn. You will be asked questions **on Lab 1 and Lab 2 only**. The Examiner may even ask questions not directly linked to what you have done to test your **understanding**. In addition to understanding, examiner will also consider the effort you have put into completing the experiments.

**Grading** – Attached is the marksheet that your Examiner will be using during the oral examination. The assessment form with feedback will be returned to you via MS Teams. Your will receive a grade for this assessment according to the criteria shown below:

| | |
|---|---|
| A = Excellent (70%+) | Completed and understood everything with no gap in knowledge |
| B = Good (60%+) | Completed all and good understanding with minor gaps |
| C = OK (50%+) | Fair understanding with some gaps, completed most parts |
| D = Weak (40%+) | Weak understanding with many gaps |
| E = Fail (below 40%). | Very weak understanding, lack evidence of real effort |

We expect the average mark for this assessment to be around 65% for the entire class. To ensure fair assessment, I will perform moderation of marks between different Examiners and apply adjustments if necessary.

**Learning Outcomes – Lab 1 and Lab 2**

## Lab 1 – Amplification

- Characterisation of signal sources with understanding of source impedance and loading effect
- Building of working op-amp circuits in a tidy manner
- Isolating load from source using op-amp using unity gain buffer
- Shifting reference voltage for single power supply operation
- Biasing input and output at middle of supply range
- AC vs DC coupling
- Gain-Bandwidth limitation of op-amp
- Interpretation of op-amp datasheet
- Multi-stage amplification
- Driving of very low output impedance (e.g. speaker) via class D amplifier

## Lab 2 – Applications of Op-amps

- Rectifier circuit
- Peak detector and how to choose R C values of the detector
- Voltage comparator with hysteresis and what determines the hysteresis
- Oscillator with positive feedback, and what determines the frequency of oscillation
- Function generator and how it works
- Analogue pulse-width modulator and how it works
- Active filter using Sallen-Key topology
- Butterworth filter design

Peter Cheung

Version 2.1, 24 October 2026.

# Mid-term Lab Oral Schedule – 2/3 November 2026

| First Name | Surname | Date | Time | Examiner |
|---|---|---|---|---|
| Raphael | Morris | Monday | 09.00 - 09.15 | Examiner_1 |
| Zaid | Haris | Monday | 09.00 - 09.15 | Examiner_2 |
| Ali | Ahmed | Monday | 09.00 - 09.15 | Examiner_3 |
| Shreya | Rao | Monday | 09.15 - 09.30 | Examiner_1 |
| Mann | Ng | Monday | 09.15 - 09.30 | Examiner_2 |
| Yiding Daniel | Zheng | Monday | 09.15 - 09.30 | Examiner_3 |
| Katherine | Chahbani | Monday | 09.30 - 09.45 | Examiner_1 |
| Aryaman | Allam | Monday | 09.30 - 09.45 | Examiner_2 |
| Jiexiang | Lei | Monday | 09.30 - 09.45 | Examiner_3 |
| Justin | Luo | Monday | 09.30 - 09.45 | Examiner_4 |
| Shaun | Sundralinghar | Monday | 09.30 - 09.45 | Examiner_5 |
| Rory | Jones | Monday | 09.45 - 10.00 | Examiner_1 |
| Pratik | Dhar | Monday | 09.45 - 10.00 | Examiner_2 |
| Xizhe | Xu | Monday | 09.45 - 10.00 | Examiner_3 |
| Beiyan | Shi | Monday | 09.45 - 10.00 | Examiner_4 |
| Shihan | Tan | Monday | 09.45 - 10.00 | Examiner_5 |
| Zafir | Ahsan | Monday | 10.00 - 10.15 | Examiner_1 |
| Hywel | Evans | Monday | 10.00 - 10.15 | Examiner_2 |
| Kerem | Sever | Monday | 10.00 - 10.15 | Examiner_3 |
| Diya | Palanna | Monday | 10.00 - 10.15 | Examiner_4 |
| Jayden Wen M | Lim | Monday | 10.00 - 10.15 | Examiner_5 |
| Devesh | Khemani | Monday | 10.15 - 10.30 | Examiner_1 |
| Charlie | Roberts | Monday | 10.15 - 10.30 | Examiner_2 |
| Yaseen | Zoarder | Monday | 10.15 - 10.30 | Examiner_3 |
| Atchuthan | Senthilkumar | Monday | 10.15 - 10.30 | Examiner_4 |
| Pin Xie | Loo | Monday | 10.15 - 10.30 | Examiner_5 |
| Ohm | Laisathit | Monday | 10.45 - 11.00 | Examiner_1 |
| Han-Yi | Ni | Monday | 10.45 - 11.00 | Examiner_2 |
| Anthony | Kommatas | Monday | 10.45 - 11.00 | Examiner_3 |
| Sarthak | Agarwal | Monday | 10.45 - 11.00 | Examiner_4 |
| Zhengxin | Cai | Monday | 10.45 - 11.00 | Examiner_5 |
| Bryan | Liow | Monday | 11.00 - 11.15 | Examiner_1 |
| FONG WAI | WONG | Monday | 11.00 - 11.15 | Examiner_2 |
| Ananthanaray | Nair | Monday | 11.00 - 11.15 | Examiner_3 |
| Ayenoya | Alegeh | Monday | 11.00 - 11.15 | Examiner_4 |
| Seungwon | Na | Monday | 11.00 - 11.15 | Examiner_5 |
| Tibor | Varga | Monday | 11.15 - 11.30 | Examiner_1 |
| Zichun | Jin | Monday | 11.15 - 11.30 | Examiner_2 |
| Xizhe | Xu | Monday | 11.15 - 11.30 | Examiner_3 |
| Manthan | Dave | Monday | 11.15 - 11.30 | Examiner_4 |
| Henry | Mak | Monday | 11.15 - 11.30 | Examiner_5 |
| Tunir | Sarkar | Monday | 11.30 - 11.45 | Examiner_1 |
| Zhuzhuochen | Ren | Monday | 11.30 - 11.45 | Examiner_2 |
| Jiexiang | Lei | Monday | 11.30 - 11.45 | Examiner_3 |
| Syed Ahmed | Iqbal | Monday | 11.30 - 11.45 | Examiner_4 |
| Yinuo | Huang | Monday | 11.30 - 11.45 | Examiner_5 |
| Adam | Adamczak | Monday | 11.45 - 12.00 | Examiner_1 |
| Rafael | Alves | Monday | 11.45 - 12.00 | Examiner_2 |
| Morgan | Bleasdale | Monday | 11.45 - 12.00 | Examiner_3 |
| Fred | Ayres | Monday | 11.45 - 12.00 | Examiner_4 |
| Eren | Kusku | Monday | 12.00 - 12.15 | Examiner_1 |
| Alexander | Simpson | Monday | 12.00 - 12.15 | Examiner_2 |
| Jack | Ward | Monday | 12.00 - 12.15 | Examiner_3 |
| Colin | Qian | **Tuesday** | 12.00 - 12.15 | Examiner_4 |

| Gp No | First Name | Surname | Date | Time | Examiner |
|---|---|---|---|---|---|
| 22 | Sami | Benhadj Ta | Tuesday | 09.00 - 09.15 | Examiner_1 |
| 31 | Hashim | Jastaniah | Tuesday | 09.00 - 09.15 | Examiner_2 |
| 18 | Leal | Bodemeaic | Tuesday | 09.00 - 09.15 | Examiner_3 |
| 21 | Hanting (Justi | Huang | Tuesday | 09.00 - 09.15 | Examiner_4 |
| 22 | Fudail | Fauzy Omar Ba | Tuesday | 09.15 - 09.30 | Examiner_1 |
| 31 | Seb | Leeming-P | Tuesday | 09.15 - 09.30 | Examiner_2 |
| 18 | Ian | Tan | Tuesday | 09.15 - 09.30 | Examiner_3 |
| 21 | Zhihan (Andy) | Li | Tuesday | 09.15 - 09.30 | Examiner_4 |
| 40 | James | Perillo | Tuesday | 09.30 - 09.45 | Examiner_1 |
| 14 | Suki | Fong | Tuesday | 09.30 - 09.45 | Examiner_2 |
| 20 | Ahmed | Okesola | Tuesday | 09.30 - 09.45 | Examiner_3 |
| 23 | Alexander | Lodge | Tuesday | 09.30 - 09.45 | Examiner_4 |
| 40 | Vaishnavi | Rajput | Tuesday | 09.45 - 10.00 | Examiner_1 |
| 14 | Alex | Gray | Tuesday | 09.45 - 10.00 | Examiner_2 |
| 20 | Sahil | Zaki | Tuesday | 09.45 - 10.00 | Examiner_3 |
| 23 | Hosam | Mohamed | Tuesday | 09.45 - 10.00 | Examiner_4 |
| 46 | Oluwadamilo | Omorodion-I | Tuesday | 10.00 - 10.15 | Examiner_1 |
| 9 | Waad | Abdulla | Tuesday | 10.00 - 10.15 | Examiner_2 |
| 4 | Yuxuan | Huang | Tuesday | 10.00 - 10.15 | Examiner_3 |
| 7 | Srihari | Karri | Tuesday | 10.00 - 10.15 | Examiner_4 |
| 46 | Zoe | Yoong | Tuesday | 10.15 - 10.30 | Examiner_1 |
| 9 | Bruna | Silva | Tuesday | 10.15 - 10.30 | Examiner_2 |
| 4 | Christoper Jie | Koh | Tuesday | 10.15 - 10.30 | Examiner_3 |
| 7 | Finn | Pendlebur | Tuesday | 10.15 - 10.30 | Examiner_4 |
| 36 | Mishti | Busa | Tuesday | 10.45 - 11.00 | Examiner_1 |
| 25 | Avigat | Gupta | Tuesday | 10.45 - 11.00 | Examiner_2 |
| 15 | Suzanna | MacAlister | Tuesday | 10.45 - 11.00 | Examiner_3 |
| 13 | Mariam | Durrani | Tuesday | 10.45 - 11.00 | Examiner_4 |
| 36 | Amina | Mammadz | Tuesday | 11.00 - 11.15 | Examiner_1 |
| 25 | Yusuf | Yilmaz | Tuesday | 11.00 - 11.15 | Examiner_2 |
| 15 | Aditya | Shivam | Tuesday | 11.00 - 11.15 | Examiner_3 |
| 13 | Oliver | Ho | Tuesday | 11.00 - 11.15 | Examiner_4 |
| 10 | Fahad | Khan | Tuesday | 11.15 - 11.30 | Examiner_1 |
| 39 | Junchen | Jiang | Tuesday | 11.15 - 11.30 | Examiner_2 |
| 37 | Junxiao | Liang | Tuesday | 11.15 - 11.30 | Examiner_3 |
| 19 | Aziz | Al-Kubaisi | Tuesday | 11.15 - 11.30 | Examiner_4 |
| 10 | Asim | Ozdemir | Tuesday | 11.30 - 11.45 | Examiner_1 |
| 39 | Fong | Si | Tuesday | 11.30 - 11.45 | Examiner_2 |
| 37 | Jiming | Shao | Tuesday | 11.30 - 11.45 | Examiner_3 |
| 19 | Soph | Garrett-Ril | Tuesday | 11.30 - 11.45 | Examiner_4 |

Imperial College London

**Department of Electrical & Electronic Engineering**

**EE2 Circuits and Systems Module**
**Mid-term Lab Oral**

Name of Student: _____

Names of Assessors: _____ Date: Mon / Tue

**Performance on the Lab Experiments**

**1. Logbook Quality and Effectiveness**

Highly effective | Effective | OK | Weak | Poor

**2. Ability to answer questions from the logbook**

Excellent | Good | OK | Poor | Very poor

**3. Effort to completing Lab 1 & 2**

| | | | | |
|---|---|---|---|---|
| Fully engaged Strong evidence | Good engagement Good evidence | Acceptable Engagement | Below expected Engagement | V. poor Engagement |

**Understanding and Learning Outcomes**

**4. Explanation on theories behind experiments**

Excellent | Good | OK | Poor | Very poor

**5. Examiner's opinion on candidate's depth of understanding in general**

Broad & deep | Good | Average | Less than average | Poor

**FEEDBACK TO STUDENT:** **Grade:**